Quiz 8

Friday, October 30 — systems, elimination, and elementary matrices

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Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Covers: solving linear systems by elimination and elementary matrices (Lecture 21, Lecture 22, Lab 2 / RREF practice). Same-day powers and Lab 3 are **not** on this quiz.

Schedule

Elementary-matrix templates (also provided on the quiz paper) are at the end of this page.

Core

Part A — Augmented matrix, eliminate, check

For each system: (i) write the augmented matrix $[A \mid b]$; (ii) solve by elimination (row operations on the augmented matrix); (iii) check by substituting your solution into the original equations (or computing Ax and comparing to b).

C8.1. (2×2)

$$\begin{aligned} x_1 + 2x_2 &= 5 \\ 3x_1 - x_2 &= 1. \end{aligned}$$

C8.2. $(3 \times 2$ — three equations, two unknowns)

$$\begin{aligned} x_1 + x_2 &= 3 \\ 2x_1 - x_2 &= 0 \\ 3x_1 &= 3. \end{aligned}$$

C8.3. (2×3 — two equations, three unknowns)

$$x_1 + 2x_2 + x_3 = 4$$
$$2x_1 + 4x_2 + 2x_3 = 8.$$

If there are infinitely many solutions, introduce a parameter t and write the general solution.

Part B — Name the elementary matrix

Use the templates at the end of the quiz (they will be provided). For each stated row operation on a system with the given number of equations, write the corresponding elementary matrix E.

C8.4. Two equations: add -3 times row 1 to row 2.

C8.5. Two equations: swap row 1 and row 2.

C8.6. Three equations: multiply row 2 by $1/2$.

C8.7. Three equations: add 2 times row 1 to row 3.

Part C — Apply E and rewrite the system

C8.8. Let

$$[A \mid b] = \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 2 & 1 & 0 \end{array}\right], \qquad E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}.$$

(a) Compute $E\,[A \mid b]$ by hand.
(b) Rewrite the result as a system of equations.

C8.9. Let

$$[A \mid b] = \left[\begin{array}{ccc|c} 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 3 \\ 2 & 0 & 1 & 1 \end{array}\right], \qquad E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}.$$

(a) Compute $E\,[A \mid b]$ by hand.
(b) Rewrite the result as a system of equations in x_1, x_2, x_3.

Stretch

S8.1. Continue C8.8: from your new matrix, finish elimination and solve the 2×2 system. Check $Ax = b$.

S8.2. For C8.3, write one particular solution (a specific numerical vector) and one nontrivial nullspace direction (homogeneous solution).

What a quiz item might change

The right-hand side, which row operation is named, or whether the system is 2×2, 3×2, or 2×3.

Elementary-matrix templates (provided on the quiz)

Left-multiplying $[A \mid b]$ by E performs the row operation. Build E by doing that same operation to the identity.

Two equations (2×2 templates):

Swap the two rows:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Multiply row 1 by $c \neq 0$, or multiply row 2 by $c \neq 0$:

$$\begin{bmatrix} c & 0 \\ 0 & 1 \end{bmatrix}, \qquad \begin{bmatrix} 1 & 0 \\ 0 & c \end{bmatrix}.$$

Add c times row 1 to row 2, or add c times row 2 to row 1:

$$\begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix}, \qquad \begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix}.$$

Three equations: start from I_3 and edit the same spot. Examples:

Multiply row 2 by $c \neq 0$:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & c & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Add c times row 1 to row 3:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ c & 0 & 1 \end{bmatrix}.$$

(The extra number sits in the row you change, in the column of the row you copy from.)